EE4-05
EE9-SO7
EE9-FPN2-01

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2019

MSc and EEE PART IV: MEng and ACGI

Corrected Copy

TRAFFIC THEORY & QUEUEING SYSTEMS

Monday, 20 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : J.A. Barria
Second Marker(s) : D.P. Mandic

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Special information for students

1. Erlang Loss formula recursive evaluation:

$$E_N(\rho) = \frac{\rho E_{N-1}(\rho)}{N + \rho E_{N-1}(\rho)}$$
$$E_0(\rho) = 1$$

2. Engset Loss formula recursive evaluation (for a fixed M and $p = \alpha/1+\alpha$):

$$e_N = \frac{(M-N+1)\alpha e_{N-1}}{N + (M-N+1)\alpha e_{N-1}}$$
$$e_0 = 1$$
$$\alpha = \lambda/\mu$$

3. Traffic capacity on basis of Erlang B formula (next page).

4. Expected residual time

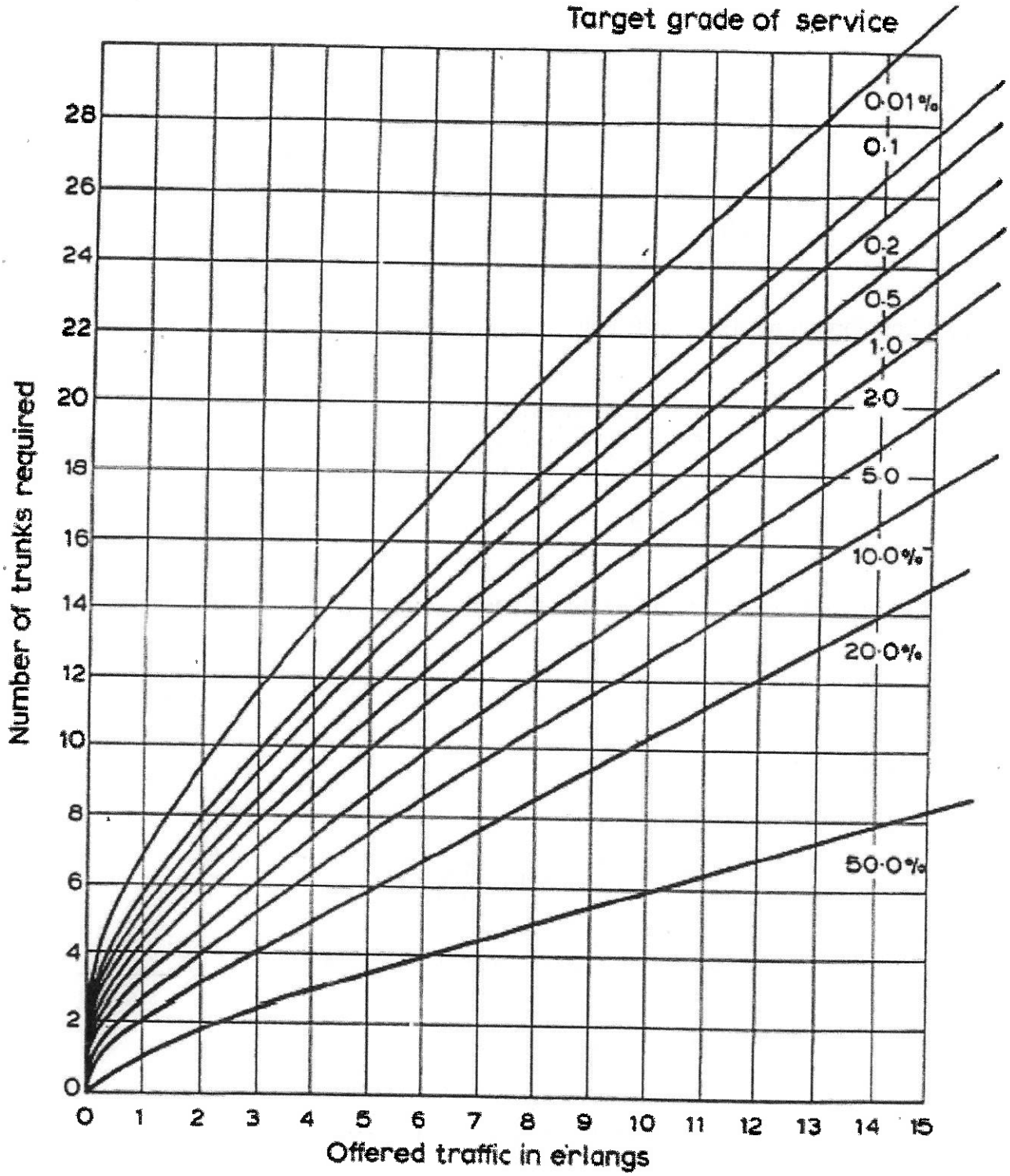
$$E[R] = \frac{1}{2}\lambda E[S^2] = \frac{1}{2}\sum_{k=1}^{m}\lambda_k E[S_k^2]$$

5. Mean and variance of the Kosten overflow model (infinite overflow link $N = \infty$).

$$m = E[N_t] = \rho E_M(\rho)$$

$$v = Var(N_t) = m\left(1 - m + \frac{\rho}{M+m+1-\rho}\right)$$

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Traffic capacity on basis of Erlang B. formula.

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The Questions

1.

a) Engset model.

i) Describe and discuss the underlying assumption of the Engset model. [2]

ii) Derive the coefficients of the Engset model. [2]

iii) Discuss under which conditions the Engset model reverts to the Erlang model, and hence highlight the underlying assumptions of the Erlang model. [2]

b) Consider an underlying discrete-state, continuous time Markov chain $\{N_t\}$ with the state space: $E = \{1, 2, 3, \ldots, N\}$ representing the Erlang model.

i) For the Erlang loss model, derive the mean carried traffic, $E[N_t]$. [6]

ii) For the Erlang loss model show, step by step, that the variance of the carried traffic can be represented by the following expression: [8]

$$Var(N_t) = E[N_t] - \rho B_c(N - E[N_t])$$

Where ρ = is the mean offered traffic and B_c= is the probability that a call is blocked.

2.

a) *Figure 2.1.* represents a $2-D$ birth and death process, $\{(M_t, N_t)\}$, with the state space $E = \{(i,j)|0 \leq i \leq M, 0 \leq j \leq N\}$, and assuming that the offered traffic is pure chance traffic with parameters (λ, μ):

i) Assign and complete all missing transition rates in *Fig. 2.1.* [6]

ii) Identify which type of traffic model the underlying $2-D$ Markov chain is representing [2]

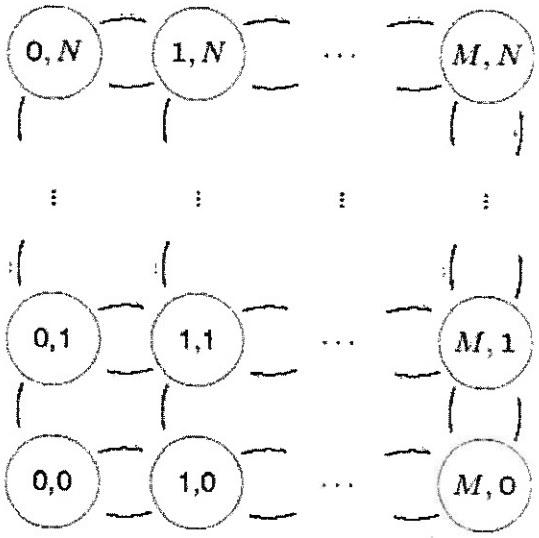


Figure 2.1. 2-D Markov chain.

b) For an $M/G/1$ queueing system with the pre-emptive resume priority mechanism:

i) Sketch the time sequence of a class k arrival and departure to a pre-emptive resume priority mechanism. [2]

ii) Identify all the random variables in your sketch. [2]

iii) Show that for a pre-emptive resume priority mechanism, the effective service time for a class k customer is scaled by the factor $(1-\sigma_{k-1})^{-1}$, where $\sigma_{k-1} = \sum_{i=1}^{k-1} \rho_i$. And ρ_i is the offered traffic in class i. [8]

3.

a) A 15-channel communications link is offered pure chance traffic with and arrival rate of 180 call/hour and a mean holding time of 180 seconds.

Blocked demands are allowed to overflow on to a single-channel overflow link.

i) Determine the probability that the overflow channel will be blocked. [4]

ii) Construct an interrupted Poisson process (IPP) model for the overflow link using matching moments. [10]

Hint: For an IPP Markov chain model ($N = \infty$):

$$\mu(k) = \left[\frac{\alpha(\alpha+1)\dots(\alpha+k-1)}{(\alpha+\beta)(\alpha+\beta+1)\dots(\alpha+\beta+k-1)}\right]\rho^k$$

for $k = 1, 2, 3, \dots$ where

$$\mu(k) = E[\, N_t(N_t-1), \dots, (N_t-k+1)]$$

is the $k-th$ factorial moment.

b) For an M/M/K system, show that for delay arrivals, the waiting time is exponentially distributed with the mean = $[K\mu(1-\rho)]^{-1}$. [6]

4.

a) Consider a FIFO queueing systems with the total capacity of four (4) customers being either served or in the queue.

Customers arrive according to a Poisson process, with parameter λ, and the service time is exponentially distributed with mean sojourn time of $1/\mu$.

The system is composed of two servers: one server is on standby and will only be active if the queue length is greater or equal than two (2).

Assuming that the state of the system can be represented by (i, j), where i is the number of customers in the system and j is the number of active servers:

i) Identify the state space of the system. [2]

ii) Sketch the underlying state transition diagram, clearly identifying all the transition rates. [6]

iii) A reward structure is to be added that reflects the number of active servers times the number of customers in the system. Determine the value of all the elements of the reward vector. [2]

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4.

b) Consider a large number N of multiplexed ON-OFF traffic sources offered to a network access link with capacity, C. If the system is bufferless, a measure of loss is given by the probability that the input rate is greater than the access link capacity, C, that is,

$$\varepsilon = \sum_{i=J_o}^{N} \pi_i$$

where J_o is the first state where the input rate is greater than the link capacity, C.

Assume that the sources of traffic are quite bursty, that is, $N \gg 1$ and $p \ll 1$. Then the value of ε can be approximated closely by a normal distribution with the mean $m = Np$ and variance $\sigma^2 = Np(1-p)$.

i) State the continuous r.v. equivalent of ε based on the Normal distribution. [4]

ii) After replacing J_o by C, show that ε has an approximate solution given by:

$$\varepsilon = \frac{\sigma e^{-(C-m)^2/2\sigma^2}}{\sqrt{2\pi}(C-m)}$$

[6]

Assume that: $\frac{(C-m)}{\sqrt{2}\sigma} > 3$

Hint:

$$\int_x^{\infty} e^{-y^2} dy = \frac{e^{-x^2}}{2x} - \int_x^{\infty} \frac{e^{-y^2}}{2y^2} dy$$